Green Bond Policies Driving Circular Economy Goals: Potential Gains and Roadblocks

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ABSTRACT

The fast-paced growth of green bonds has positioned them as an important instrument for financing the transition toward a circular economy (CE). However, empirical evidence on how green bond proceeds are allocated and aligned with CE principles in emerging economies remains limited. This study examines green-bond–financed projects under Series VII A and VII B of the Indian Renewable Energy Development Agency (IREDA) using structured content analysis of ESG reports, annual reports, and sustainability disclosures. The findings show that green bonds effectively mobilize capital for renewable-energy–based CE initiatives, particularly those related to resource efficiency, waste reduction, and sustainable infrastructure. Nevertheless, challenges persist due to fragmented reporting practices, greenwashing concerns, and regulatory inconsistencies. The analysis further indicates that firms with transparent fund allocation strategies enhance investor trust, thereby supporting broader market participation. By providing project-level evidence from India, this study offers policy and investment insights for strengthening the role of green bonds in advancing circular economy transitions.

Keywords: Green bonds, Circular Economy, Sustainability, Sustainability Reporting, IREDA, Green Finance, India

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INTRODUCTION

A shift to a circular economy (CE)[2] demands substantial financial resources, even as traditional linear models remain dominant in global production and consumption systems. Green bonds are emerging as a pivotal mechanism to bridge this financing gap, offering issuers access to capital while enabling investors to support projects aiming for sustainability (International Capital Market Association, 2021). Unlike conventional bonds, green bonds are explicitly tied to financing activities like deployment of renewable energy, sustainable infrastructure, eco-friendly transportation, and waste management (EUROSIF, 2018). Their dual value lies in combining fiscal returns with measurable environmental impact.

Even though green bonds are gaining importance, research based on the intersection of green bonds and the adoption of CE remains fragmented. Prior studies have primarily focused on the green bond market and its evolution, investor motivations, and its contribution to climate finance (Banga, 2018; Tolliver et al., 2020). However, less is known about how green bond proceeds are allocated at the project level and whether these allocations align with CE principles in practice. This gap is quite relevant for emerging economies like India, where climate vulnerability and ambitious sustainability commitments — such as accomplishing a target of net zero emissions by 2070 (Government of India, 2022) — require innovative financing mechanisms.

This study addresses this gap by examining projects financed under Series VII A and VII B of the IREDA. Through a qualitative content analysis of organisational ESG and sustainability documents, the study explores the extent to which green bond funding supports CE initiatives, while also identifying persistent financial, operational, and regulatory challenges. By doing so, the paper makes a contribution to the green finance literature based on evidence from India and provides implications for policymakers, companies, and investors looking to utilize green bonds as a catalyst for promoting CE transitions.

LITERATURE REVIEW

Interest in the circular economy (CE) has increased substantially over the past few years, leading to higher participation in spite of the continued problem in accurately qualifying the concept. The natural lack of clarity over CE has enabled the application of new and unconventional methods, aligned with theoretical approaches associated with the diffusion of innovation theory (Di Vaio et al., 2023). The Ellen MacArthur Foundation (2013) states that a CE can be defined as a regenerative and restorative system which keeps materials in their finest utility and value. The key business principles that form the basis of CE practices are maximizing product durability through maintenance, refurbishment, and remanufacturing, and recycling strategically to reuse high-quality recyclables (Milios, 2021). A CE positively impacts local employment through developing skilled expertise in remanufacturing and restoring used items (Rosa et al., 2019). Shifting from conventional production practices towards cyclical systems can lead to economic, environmental, and societal advantages (D'Adamo, 2019). The growing repository of scholarly studies and rising business involvement further enhance the prominence of CE (Taddei et al., 2022).

However, CE projects still lag behind, considered to be high-risk activities even when there are potential gains. An apt example is that of the BlackRock Circular Economy Fund, which has been losing out since 2021, even with net assets worth US$1.7 billion. While billions are invested in circular solutions, trillions go into linear models, motivated by greater short-term profitability, slowing the transformation to a circular economy (Li et al., 2023; Patterson & Ramkumar, 2021).

Green finance (GF) plays a significant role in advancing sustainable development and achieving key environmental objectives. However, its scope remains debated: some scholars treat GF as interchangeable with broader frameworks like sustainable or ESG-aligned finance, while others argue for its distinct conceptualization as a narrower, environmentally-focused category (Hafner et al., 2020; Migliorelli & Dessertine, 2019; Lindenberg, 2014; UNEP, 2016). Green Finance has been proven to play a role in channelizing funds from different industries towards United Nations' Sustainable Development Goals (SDGs; Ahmad et al., 2022; Campiglio, 2016). A holistic approach for an integrated strategy to GF is essential for addressing climate change and enhancing resilience simultaneously. This can be done by offering various green financial products like savings, loans, insurance, and even digital payment tools through financial institutions (Desalegn & Tangl, 2022).

The attainment of the Sustainable Development Goals (SDGs) specifically Goals 1, 7, and 13, requires a universal change of investment strategy to sustainability (Ahmad et al., 2022). One important aspect of external financing is bank lending, which supports environmentally sustainable businesses (Chen et al., 2022). Other forms of financing do not compare to the empirical evidence that bank lending is the most favourable (Hussein & Hamdan, 2020). Green finance is more widely discussed, but barriers to the closing of the investment gap still exist (Kumar et al., 2022). The promotion of sustainable economic growth links CE to environmental and green finance. While financial support lags far behind what is needed, it remains the most critical asset needed to achieve the long-term benefits of CE. Adequate funding of environmental projects is crucial to overcome the effects of insufficient investment.

The World Bank spearheaded the introduction of the inaugural green bond in 2008, driven by demands from Swedish pension funds for climate-oriented investment products, thereby transforming the realm of sustainable finance. This development led to the creation of a readily tradable, fixed-income instrument, specifically targeted for climate-related initiatives, and established a foundational framework for the current green bond market. The World Bank's formal issuance practices have become best practices globally, widely codified as per the framework of the Green Bond Principles (International Finance Corporation, 2019).

REGIONAL PERSPECTIVES OF GREEN BOND MARKETS

2.1.1 China

China's green bond market is currently defined by a top-down regulatory structure in which public institutions lead growth. The model continues to widen the gap for growth, yet regulatory evasion is an issue that can be resolved with the creation of combined and unified regulatory bodies with the application of strict control policies (Li et al., 2023). The market, too, has promoted ecological innovation in companies, facilitating the softening of financing bottlenecks (Lian et al., 2024).

2.1.2 Africa

Africa presents significant untapped opportunity for the application of green bonds in climate adaptation and mitigation efforts. Kenya, Nigeria, and South Africa plays critical role. Considering the strong institutional frameworks, coordinating policies, and building partnerships between public and private sectors is also crucial (Ojo et al., 2022).

Despite of the progress, the market hardly manages with issues of under-development of project pipelines and low investor awareness.

2.1.3 Southeast Asia

In Southeast Asia, green bonds are pictured as catalysts for renewable energy development, with such initiatives being directly linked with regional business and coordination processes. To enable market development, institutionalizing standardized green bond processes, encouraging collective green technology networks, and facilitating digital exchange platforms are recommended (Nguyen & Tran, 2023). Additionally, successful navigation of regulatory hurdles requires concerted efforts from government and private sectors.

2.1.4 Latin America and the Caribbean

Green bonds in the region are influenced by various elements such as the size of issues, issuers' categorization, and currency denomination. The most significant set of challenges derives from the difficult process of the green bonds issuance and the shortage of fiscal incentives. A high-priority attention to the streamlining of issuance procedures and strengthening investor confidence is required in order to address the same (Ketterer et al., 2019).

2.1.5 South Africa

South African green bonds are instrumental to the nation's energy transition, but market growth is constrained by the sluggish implementation pace and the lack of adequate financial incentives. The introduction of guarantees, funding support, and flexible regulatory regimes would speed up additional market growth (Dave & Akongwale, 2024).

2.1.6 Canada

In Canada, the popularity of green bonds is growing, driven by increasing demand for responsible environmental investments and resulting positive reputation effects. However, hindering further growth are significant transaction costs, relatively low market liquidity, and a lack of adequate evaluation of the environmental impacts of transactions. Specific policy interventions are necessary to promote the growth of green bonds in the country (Weber & Saravade, 2019). The European Union’s green bond market has a strong profile, with its performance being positively impacted by the stability of the emissions trading system. Environmental value is represented by green bonds in that they help decrease ecological harm during times of uncertainty in the market (Leitão et al., 2021).

These regional experiences highlight the fact that while green bond markets are expanding globally, differences in regulatory maturity and reporting practices significantly influence their contribution to circular economy outcomes.

2.2 Green Bonds in India

A World Bank report (2023) indicated that green bonds have raised US$2.5 trillion universally to support sustainable and green projects. In the initial months of 2023, the Indian government successfully launched its inaugural green bond initiative, securing approximately US$2 billion in funding to support environmentally focused endeavours, with a basic aim of addressing the undesirable effects of climate change, promoting sustainability, and mitigating its adverse impacts on biodiversity and ecosystems.

Due to its pronounced vulnerability to climate change, India has set ambitious environmental goals — achieve net-zero emissions by 2070 and reduce GDP emissions intensity by 45 percent from the 2005 levels by 2030 (Government of

India, 2022). To accomplish these objectives, substantial financial resources are required, particularly in relation to investments in clean technologies and sustainable infrastructure. It is estimated that to achieve net-zero emissions by 2050, an investment of US$12.7 trillion is required, corresponding to an annual obligation exceeding US$470 billion (BloombergNEF, 2023). Thus, the financial requirements for climate change mitigation in India necessitate coordination of private sector finance with public sector funding.

The green bond market shows considerable promise as a mechanism for garnering private investment, driven by its application in environmental projects (Agarwal & Singh, 2018). The market has shown significant growth since European Investment Bank issued the first climate-related bond in 2007. In 2023, green bond issuances amassed a total value of about US$2.334 trillion, with an increase of US$420.6 billion registered for 2023 (BloombergNEF, 2023). Green bonds, which have received substantial international acceptance, do not constitute a significant share of the Indian bond market. By the end of February 2023, India's green bond issuances reached approximately US$21 billion, which is merely about 3.8 percent of the country's domestic corporate bond market, valued at over US $500 billion. Nonetheless, the current trend in corporate social responsibility (CSR) funding towards environmentally orientated initiatives suggests a latent capacity for growth within this sector.

Weber & Sarvade (2009) point out that encouraging the market for green bonds needs implementation of tailored policy and market initiatives. The current literature indicates that institutional investors, like pension funds, insurance firms, and mutual funds, are best placed to take advantage of the long-term value of green bonds.

Green bonds have aided in climate change mitigation efforts with different degrees of application in the world. Notably, there is a growing trend in the green bond market in the developed countries as well as others that are still emerging. The rise is explained by the fact that more investors are recognizing climate change and enabling regulatory environments (Banga, 2018). On the contrary, in certain developing nations, the green bond market is not well developed; there are major challenges that deter its complete exploitation.

Monies generated by the sale of green bonds, a form of fixed income securities, are invested in financing or refinancing eligible green projects. This must be in accordance with the four fundamental Green Bond Principles (International Capital Market Association, 2022). Green Bond Principles (GBPs) are guidelines for the establishment of the green bond market. They emphasize honesty, disclosure and transparency during the issuance process. The GBPs are complex guidelines that help the issuers, investors, and underwriters comprehend the green bond market (International Finance Corporation, 2022). Issuers are required to follow four main components: (1) Use of Proceeds, in which funds must be used in only eligible green categories; (2) Project Evaluation and Selection, in which issuers develop specific eligibility criteria; (3) Management of Proceeds, where issuers generally develop tracking methods; and (4) Reporting, where issuers periodically report on the use of proceeds and its environmental impact.

The GBPs for 2025 feature enhancements. They broaden the definition of green projects to include not only the assets and expenses but also the activities involved. Likewise, they are aligned with the Green Enabling Projects Guidance, released in June 2024, that details the types of projects that qualify. Along with institutional resources, pre-issuance checklists and templates, these enhancements boost issuer transparency and stakeholder trust. Together, these crucial elements and enhancements will enable the GBPs to promote the standardization of market practices and enhance the integrity of green bond issuances.

2.3 Challenges Faced by the Green Bond Markets

The green bond market, capable of fostering sustainable development, continues to encounter various systemic challenges. Lack of standardization is a major concern, since the absence of a unified global framework and regulatory measures results in discrepancies in reporting and eligibility criteria, increasing the chances of greenwashing and investments without environmental legitimacy (Agarwal & Singh, 2018). Also, exogenous currency and exchange rate risks also narrow the market; in a country like India, without adequate stability of currency and because of the high cost of hedging, foreign investors are discouraged. Green bonds denominated in the domestic currency generally have higher yields to reflect the inherent risk (Kumar et al., 2022). Regulatory barriers also act as a hindrance to broader participation given that the current frameworks focus on investments in high-rated securities, thereby leaving out a large group of potential issuers. However, there is a promise of growth via continuous reforms, including investing pension funds in sovereign green bonds (Ojha, 2023). Liquidity constraints in the market and the lack of fit between the size of green projects and the size of bond issues favoured by funders also impede the effective mobilization of capital. Other barriers to the adoption of green bonds in emerging markets include institutional gaps in awareness and education, in particular among retail investors, and a pipeline of viable projects (World Bank, 2023; IFC & Amundi, 2024). Together, these challenges highlight the importance of strong policy initiatives, increased investor education, and harmonized frameworks in order to realize the potential of green bonds in providing finances to sustainable transitions.

MATERIALS AND METHODS

3.1 Objective and Methodology

The main aim of the research is to determine the significance of green bonds in the funding of circular economy (CE) initiatives as well as to explore the difficulty of executing green-bond-funded CE initiatives. This study employs a qualitative approach to research by conducting content analysis, as it seeks to examine the usage of green bond funding towards circular economy (CE) initiatives. The lens is on company funding through IREDA Series VII A and VII B green bond issues.

Content analysis was chosen because it was deemed to be the most appropriate tool to analyze issues. It also displays transparency and replicability in the determination of the themes of CE (Krippendorf, 2018; Neuendorf, 2017). Secondary data was sought through environmental, social, and governance (ESG) reports, annual reports, corporate sustainability reports, climate impact declarations, policy guidelines and reports on green bonds and green ceiling in the industry. The coding system was developed by relying on the four CE principles identified in the literature: optimize resources, use renewable energy, minimize waste, and create sustainable infrastructure (Ellen MacArthur Foundation, 2013; Milios, 2021). The other categories included fund distribution strategy, sustainability goals and implementation barriers.

The study focuses on IREDA Series VII A and VII B green bond issuances due to the availability of detailed, publicly accessible ESG and project-level disclosure reports. These series represent some of the most transparent public green bond issuances in India, making them suitable for an in-depth qualitative content analysis. While the scope is limited, it allows for a focused examination of project-level alignment between green bond financing and circular economy principles.

3.2 Identification of Key Themes and Indicators for Circular Economy Principles

Distinct subject areas, including "resource efficiency", "renewable energy", "waste reduction", "sustainable infrastructure”, and “sustainable infrastructure construction" were identified. Topics include "fund allocation", "sustainability goals", "deployment of circular economy policies", and "challenges ". A qualitative review of reports was conducted to extract relevant data. The frequency of corporate environmental phrases including "minimization of waste", "circular model", "carbon neutrality", and "resource efficiency” was recorded. Verification with secondary sources, like publications from the Climate Bonds Initiative, was conducted to confirm accuracy. The reliability and consistency of results were also ensured by employing triangulation techniques. Terms related to corporate sustainability (e.g., "circular economy", "carbon neutrality", and "resource efficiency") were identified and classified. Comparisons between companies revealed similarities and differences in CE usage. The findings were cross-verified by utilizing multiple sources such as publications from the Climate Bonds Initiative and global industry standards. The coding was carried out in iterations and repeated to reduce the chance of subjectivity and to increase the reliability of the results.

A deductive coding approach was adopted based on the four CE principles identified in the literature review: resource efficiency, renewable energy use, waste minimization, and sustainable infrastructure development. Relevant textual segments from corporate reports were coded iteratively, and the findings cross-validated using secondary sources such as Climate Bonds Initiative publications. The coded textual segments were further examined in detail. Statements describing the reduction of dependence on fossil fuels through rooftop or distributed solar installations were coded under renewable energy adoption and classified within the renewable energy utilization principle of the circular economy. References emphasizing optimization of land use, improvements in operational efficiency, or enhanced use of resources were coded as resource optimization and mapped to the resource efficiency dimension. Similarly, disclosures highlighting long-term infrastructure development aligned with sustainability objectives were coded as sustainable asset creation and categorized under the sustainable infrastructure principle. This approach demonstrates how qualitative textual data from corporate disclosures was systematically translated into analytical categories, ensuring transparency and consistency in the content analysis.

FINDINGS AND DISCUSSION

Tables 1 and 2 show a detailed evaluation of the energy projects financed by partially green bonds to assess how they meet circular economy goals. This evaluation considers important project details, financial spending, and the outcomes achieved. The projects focus on renewable energy, particularly in the solar and wind energy sectors. They play a crucial role in creating more sustainable energy systems by reducing dependence on fossil fuels and supporting clean energy sources. Solar energy projects are the most common, featuring a wide range of installations from small to large facilities. Wind energy projects, while fewer in number, are important for maintaining grid stability and boosting the production of renewable energy.

Table 1: Assessment of CE Principles in Green-Bond-Funded Projects

Name	CE Principle Followed	Explanation
Acme Solar Roof Top Systems Pvt. Ltd.	Resource Efficiency	Promotes decentralized energy generation, reducing reliance on non-renewable sources.
Cambridge Energy Resources (P) Ltd	Decentralization & Localized Impact	Supports small-scale energy production in remote areas, reducing energy loss in transmission.
ES Energy Private Limited	Grid Independence	Enhances energy resilience by reducing dependency on centralized grids.
Mytrah Abhinav Power Private Limited	Large-Scale Renewable Integration	Expands sustainable energy infrastructure, supporting long-term CE goals.
Rising Bhadla 1 Pvt. Ltd.	Infrastructure Reuse	Contributes to sustainable energy supply through optimized land use.
Skeiron Renewable Energy Amidyala Ltd	Renewable Energy Utilization	Wind energy as a sustainable alternative to fossil fuels, minimizing carbon emissions.
Vayu Urja Bharat Private Limited	Sustainable Energy Supply	Large-scale renewable projects reduce dependency on traditional energy sources.

Source: Authors' own

Table 2: Fund Allocation in Green-Bond-Financed Projects and its Impact on CE Initiatives

Name	*Sector*	*Capacity (MW)*	*Project Cost (INR crores)*	*Funds Allocated from Green Bonds (INR crores)*	*Industry*	*CE-Related Projects*	*Impact Metrics*	*Challenges*
Acme Solar Roof Top Systems Pvt. Ltd.	Solar	30	226.68	17.17	Renewable Energy	Rooftop solar installations.	Reduced carbon emissions; enhanced rooftop solar adoption.	Financial viability of small-scale projects.
Cambridge Energy Resources (P) Ltd	Solar	4.23	58.12	0.85	Renewable Energy	Small-scale solar projects.	Energy supply to remote regions.	Limited scalability of projects.
ES Energy Private Limited	Solar	10	71	26.63	Renewable Energy	Distributed solar generation.	Reduced grid dependency.	Managing maintenance of small installations.

ES Solar Private Limited	Solar	10	73	27.38	Renewable Energy	Distributed solar projects.	Improved solar energy availability.	Financing additional expansions.
ES Sun Power Private Limited	Solar	20	144	91	Renewable Energy	Mid-scale solar energy projects.	Energy supply for larger areas.	Dependency on seasonal solar conditions.
Harikrishnan Power And Technology Pvt. Ltd.	Solar	1	6.45	0.71	Renewable Energy	Small-scale solar installation.	Minimal environmental impact.	Limited funding for small-scale projects.
Mytrah Abhinav Power Private Limited	Solar	87	64.99	216	Renewable Energy	Large-scale solar projects.	Increased renewable energy capacity.	Ensuring ROI in large projects.
Photon Suryakiran Private Limited	Solar	70	510	2.2	Renewable Energy	Large-scale solar installations.	Sustainable energy output for large areas.	Managing high capital costs.
Rising Bhadla 1 Pvt. Ltd.	Solar	70	451.81	50	Renewable Energy	Solar park development.	Contributed to grid-level energy supply.	Limited geographical areas for expansion.
Rising Bhadla 2 Private Limited	Solar	70	451.81	61.05	Renewable Energy	Solar park development.	Significant renewable energy contribution.	Dependency on grid connectivity.
Samyama Jyothi Solar Energy Pvt. Ltd.	Solar	3	20.16	3.07	Renewable Energy	Small-scale solar installations.	Reduced carbon footprint in local areas.	High maintenance for smaller projects.

Skeiron Renewable Energy Amidyala Ltd	Wind	226.8	1788.7	74.05	Renewable Energy	Large-scale wind power generation.	Enhanced grid stability through wind energy.	Seasonal wind dependencies.
Vayu Urja Bharat Private Limited	Wind	120	1108	129.89	Renewable Energy	Large-scale wind energy generation.	Significant renewable energy contribution.	Ensuring transmission network efficiency.

Note: All these projects are partially funded by green bonds

Source: Authors' own

Green bonds have been essential in financing numerous renewable energy initiatives. For instance, the extensive solar projects supporting Rising Bhadla 2 Private Limited secured INR61.05 crores. Conversely, smaller solar projects backed by Cambridge Energy Resources obtained INR0.85 crores. This indicates that green bonds can aid both major and minor renewable energy initiatives.

These initiatives help fulfil CE goals in three primary domains. The installation of rooftop and distributed solar energy systems results in substantial decreases in carbon emissions. Small solar installations also reduce the local carbon footprint and enhance sustainability in regional energy systems. Investments in energy accessibility initiatives, such as Cambridge Energy Resources, seek to deliver renewable energy to isolated and underserved regions. This results in improved energy equity. Wind energy initiatives enhance grid reliability and facilitate seamless integration of renewable energy into grid systems. Large-scale wind projects by Skeiron Renewable Energy Amidyala Ltd. have notably increased the reliability of energy supplies at the grid level.

The execution of CE projects faces significant obstacles that prevent them from realizing their full potential. A major challenge is the significant capital investment required for large-scale projects. This poses a considerable danger to investments while also reducing the returns on investment. This financial strain is especially relevant for companies like Photon Suryakiran Private Limited. Small-scale CE projects face limitations due to restricted funding and regional circumstances. These scalability constraints hinder project expansion and increase expenses. The quantity of resources utilized by small projects, such as Cambridge Energy Resources, exemplifies these challenges. CE projects typically rely on seasonal demand, and face varying levels of solar and wind energy generation. They are susceptible to changes in the environment. The reliability of these initiatives is connected to uncertain climate factors. This is a limitation with seasonal setups in general, including the initiative by ES Sun Power Private Limited. Another challenge associated with renewable energy is insufficient infrastructure, which impedes large-scale integration and energy transmission. Innovative initiatives like the Rising Bhadla Project, a solar park endeavour, highlight that grid connectivity and transmission effectiveness are major constraints on reaching complete CE potential.

From the above discussion, it is evident that green bonds serve as an important tool in propelling CE-projects renewable energy projects. Of particular significance is the fact that the sustainable business model of Vayu Urja Bharat Private Limited has managed to achieve decarbonization and grid stability enhancement objectives using sustainable energy output systems. Although green bonds have continued to be unavoidable in the pursuit of the objectives of CE, ensuring greater project expansion, elimination of environmental degradations, and reduced dependency of its operations on unpredictable environmental interventions would enhance the effectiveness of such undertakings. In order to sustain green development, renewable energy proposals must be thought through in totality, combining innovation incentives, grid infrastructure advancement and massive investments in the sustainability and viability of the environment. The present work provides project-level evidence of the mobilization of green bond proceeds towards circular economy (CE) projects in India as applicable to firms financed as part of the IREDA ISS VII A and VII B. The findings from the study indicate that green bonds are effective in channelling funds toward renewable energy initiatives, with a considerable portion of the investments going to solar and wind energy. These initiatives enhance CE principles by maximizing resource efficiency, minimizing carbon footprints, and fostering sustainable infrastructure.

However, the examination also reveals persistent problems. Initially, financial and operational challenges — significant upfront capital demands and difficulties in scalability —impair the enduring viability of both small and large enterprises. Secondly, inconsistencies in regulations and reporting diminish transparency, reinforcing investor concerns about greenwashing. Thirdly, context-related challenges like seasonal variations in renewable resource production and deficiencies in grid connectivity impede project effectiveness.

LIMITATIONS OF THE STUDY

The current research possesses some limitations despite its benefits. Firstly, the study depends on secondary data sources, which may not fully illustrate the short-term impact of initiatives funded by green bonds. Secondly, the analysis includes only companies associated with IREDA. Consequently, its outcomes are relevant only to a particular sector. Moreover, it overlooks the sustainability of the financial aspects of these initiatives, financed primarily through green bonds, over time.

IMPLICATIONS

The findings from the study reveal that it is imperative for policymakers to amplify regulatory precision and consistency to facilitate increased participation in green bond markets. To promote the adoption of green bonds, investors and issuers should be motivated through advantageous tax treatments and mechanisms mitigating risk. In addition, fund disbursement and transparency of impact assessment must be improved to enhance investor confidence. Future research should include cross-country comparisons and the use of primary data to enhance our understanding of the contribution of green bonds to achieving CE objectives.

CONCLUSION

An extensive evaluation of some initiatives supported by the Series VII A and VII B green bonds released by the Indian Renewable Energy Development Agency has shown that green bonds play the role of a catalyst for transitioning towards circular economy. The extracted data indicates that green bonds are vital financial tools to promote a circular economic strategy, while tackling sustainability challenges across various industries. This is particularly crucial for projects involving renewable energy and sustainable infrastructure, while also tackling persistent ecological issues.

Nevertheless, considerable financial, regulatory, and operational barriers still exist for the adoption of green bonds. Initially, these difficulties must be tackled via standardized performance assessment metrics, alleviation of investor worries regarding greenwashing, and resolution of uncertainties in the regulatory landscape. Secondly, to foster trust among investors and validate the long-term viability of green-bond-funded initiatives, authorities need to enhance regulatory frameworks to guarantee clarity in the allocation of project funds, and similarly aligned expectations concerning the evaluation of sustainability performance.

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